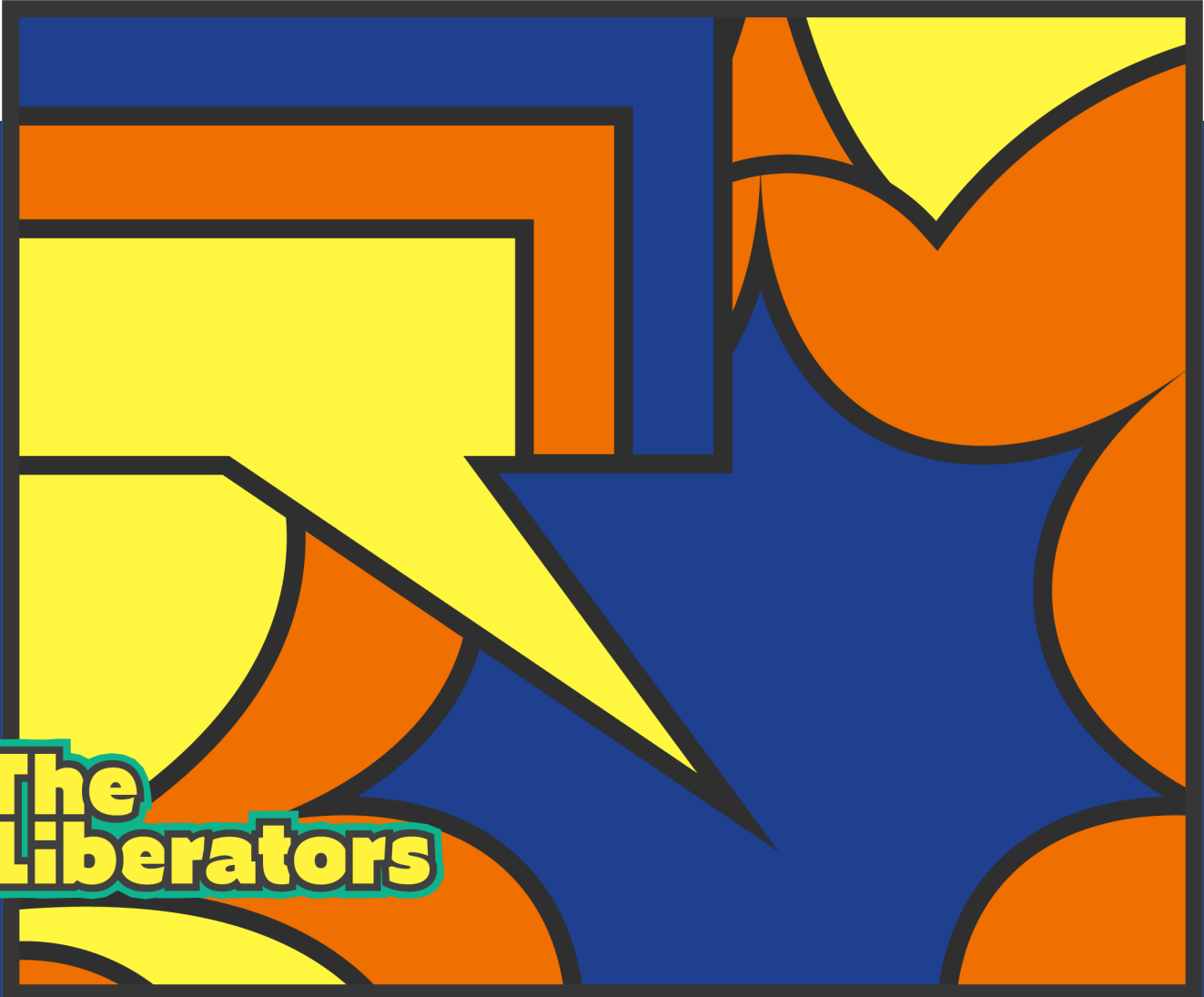


Use Critical Incident Reviews To Build Antifragility

A simple experiment to encourage self-organization and inquiry & dialogue by reflecting on critical incidents as they happen



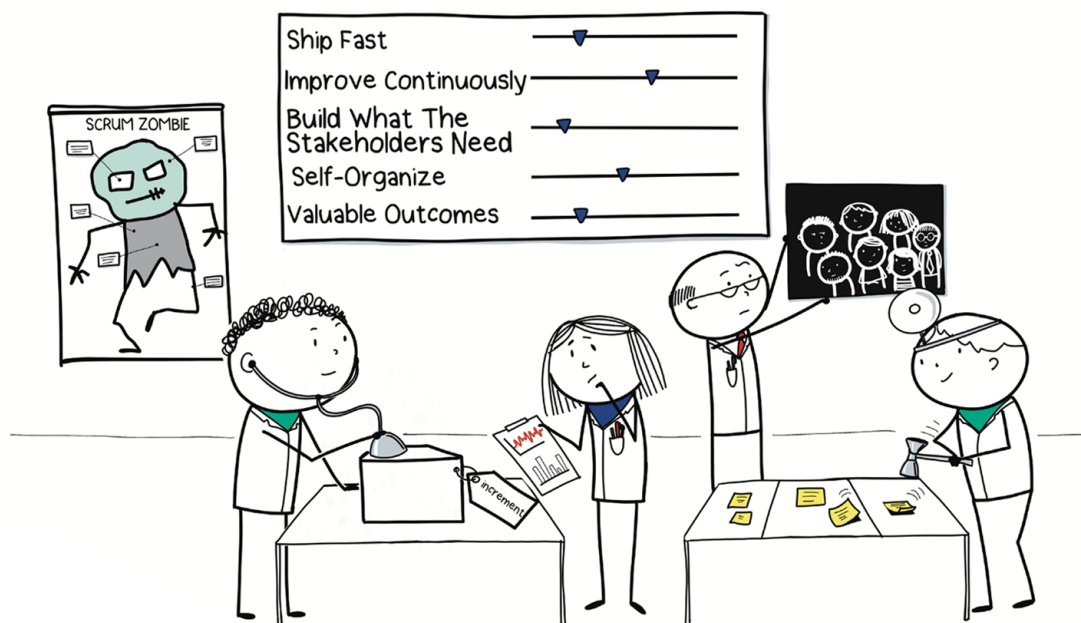
**The
Liberators**

Purpose Of This Experiment

Mistakes are inevitable in complex work. Instead of trying to avoid them, you can use them to grow stronger. Unfortunately, many Scrum teams don't know how to effectively learn from mistakes.

Yet, improving *from* mistakes is at the core of "Antifragility". Instead of trying to resist variation and shocks, antifragile systems grow stronger when they are pressured. For example, engineering teams at Netflix created a tool called "Chaos Monkey" to randomly terminate services in their infrastructure. Every time a terminated service ends up causing disruptions to end-users, engineering teams redesign the architecture to reduce the impact. Over time, responding to random shocks helps Netflix to make its infrastructure stronger. Space Exploration Technologies (SpaceX) has a launch cadence that is purposefully higher than other launch providers. Every time a launch fails, their self-managed teams update technology, protocols, and processes to avoid similar failures in the future.

This experiment uses the Liberating Structure ["What, So What, Now What"](#) to help teams reflect on failure and use it to identify what can be done to make the system (of their team & organization) stronger against similar and other failures.



Created by Thea Schukken for the Zombie Scrum Survival Guide
by Christiaan Verwijs, Johannes Scharlau & Barry Overeem
zombiescrum.org

Difficulty / Effort



Required Skill

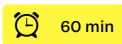


Critical Incident Reviews benefit from clear facilitation and asking powerful questions to dig deeper.

Impact On Survival



The skill to analyze and learn from mistakes is vital for self-organization and continuous improvement.



Steps



To implement this experiment, do the following:

1. Run this experiment as soon as possible after your Scrum team experiences a mistake. This can be a huge, highly impactful mistake or a smaller one. The more serious the mistake, the more time you want to take to learn from it. Include whoever was involved in, or affected by, this mistake.
2. First, give people time to detach themselves from the emotions surrounding the mistake. In random pairs, ask people to share their experience of the mistake and its consequences **2 min**. Repeat three more times in new pairs. Then ask the whole group to share two or three patterns they noticed **5 min**.
3. Set the stage of the Critical Incident Review. The purpose is to learn from a mistake and to prevent similar ones in the future. Emphasize that they are not to assign blame, even when the mistake is attributable to a single person or subgroup because those people are part of a larger group that could've helped prevent the mistake.
4. In small groups, ask everyone to retell the story of the mistake. Ask "Working backward from the moment you discovered the mistake, create a timeline of what happened. What were the actions? Who were the actors? How did information flow? Who was missing?". Encourage people to channel their inner detective and leave opinions, interpretations, and conclusions aside for now. Give everyone time to get their own thinking started **2 min**, then work together in their small groups to create the timeline **15 min**. With the whole group, share the timelines and notice similarities and differences **10 min**.
5. Now that the group has a better sense of what actually happened, ask "What is important about this? What does this mean about our work as a team? What conclusions can we draw?". First individually and in silence **2 min**, then invite people to share their ideas in their small groups **5 min**. Capture the most important insights with the whole group **10 min**.
6. Help the group turn their discoveries from the previous round into improvements. Ask "Now what? How can we reduce the blast radius of similar mistakes in the future or avoid them altogether?". First individually and in silence **2 min**, then invite people to share their ideas in their small groups **5 min**. Capture the most important ideas with the whole group **10 min**. Formulate [15% Solutions](#) for the most promising actions.



Our Findings

- For another perspective, you can use different colors for everyone in your team. This allows you to identify who is most often in need of permission.
- If you want to focus on the amount of organizational bureaucracy, don't add permission tokens for requests from direct stakeholders like customers, users, or people who otherwise invest significant money or time in your product.

Experience From The Field

"A few years ago, most of our web-based infrastructure collapsed all of a sudden. Everyone scrambled to figure out what happened while at the same answering phone calls and emails from concerned customers. It turned out that a few SSL-certificates had expired at the same time, some of which were used by web-services that most of our web-applications relied on. Although finding and fixing the issue took 25 minutes, it took much longer for us to appease angry customers.

During a 'Critical Incident Review' with the entire team, we discovered that one member of our team had been notified of the upcoming expiration a week before, but ignored the email because he assumed it had already been taken care of. Since we relied on many different SSL-certificates, we often received notifications about expiration – some important, some not. To strengthen our system against similar failures in the future, we developed three solutions.

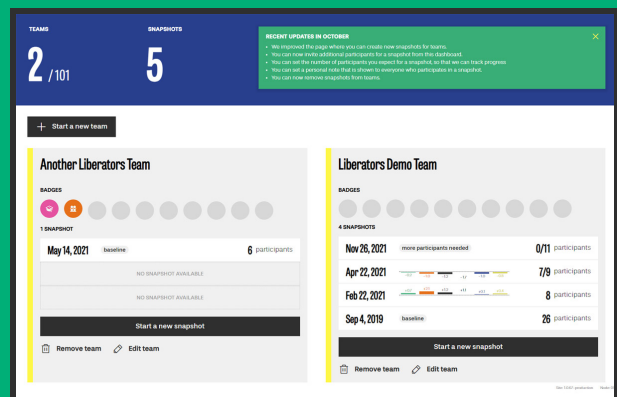
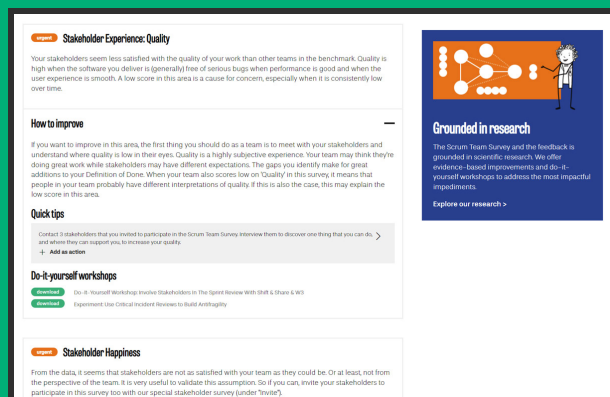
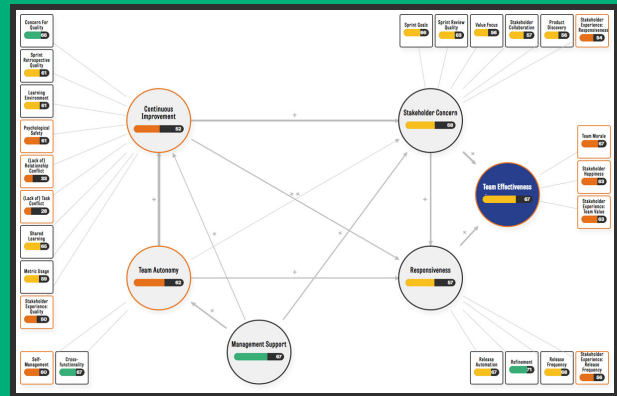
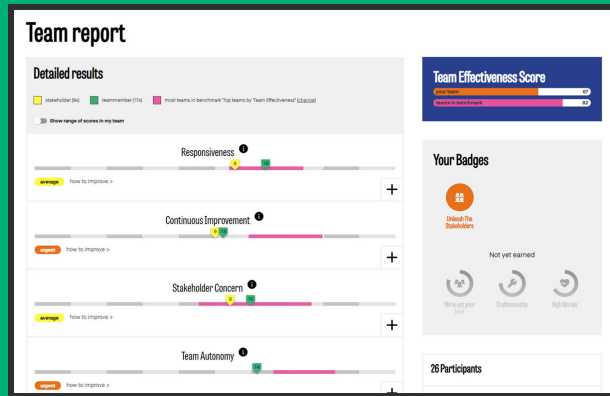
The first was the installation of a simple monitoring tool on a screen in the team room that turned red when an SSL-certificate was within days of expiration. The second was that we switched to auto-renewed SSL-certificates wherever possible, starting with the most critical services – a technology that wasn't available before. Finally, we made the responsibility to update SSL certificates (and monitor them) a responsibility of the entire team – not just one person. Since that review, we haven't had other issues with expiring SSL-certificates. In fact, we found the monitoring tool so helpful that we expanded it with checks for response-time, certain security issues, and 'chatty HTTP headers'."

More Resources

- Blogpost: [Create shared understanding with 'What, So What, Now What'](#)
- Book: [Antifragility by Nassim Taleb](#)
- Podcast: [Thinking by Sprinting](#)

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	Essential	Liberator
Invite team members & stakeholders	unlimited	unlimited
Quick Tips for your team per topic	3	unlimited
Recommendations to your team	5	unlimited
Compare your team against different benchmarks		✓
Analyze the range of scores in your team		✓
Receive our personal guidance on how to get started		✓
See where your team improves over time with Team Trends		✓
Support our academic research to help more teams		✓
Pricing per team per month	€ 0	€ 10

<https://scrumteamsurvey.org>

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